

1 0

Read the following passage.

The wall gecko is a medium-sized lizard. In an isolated habitat of southern Italy, the wall gecko shows phenotypic diversity. Scientists investigated whether disruptive selection was leading to sympatric speciation in the wall gecko.

Pale geckos live only on walls and are nocturnal (active at night). Dark geckos 5 live mainly on the dark trunks of olive trees and are diurnal (active during the day). These diurnal geckos can change skin colour when occupying different surfaces during the day.

Comparison of mitochondrial genes indicated that the diurnal geckos formed a distinct genetic group. This comparison also confirmed that all the geckos in 10 the habitat were of the same species.

The scientists used the mark-release-recapture method to estimate the size of the population of geckos in the habitat.

Use the information in the passage and your own knowledge to answer the following questions.

1 0 . 1

The wall gecko shows phenotypic diversity (lines 1–2).

Suggest **two** factors that have resulted in this phenotypic diversity.

[2 marks]

1 _____

2 _____

Question 10 continues on the next page

1	0	.	4
---	---	---	---

Explain how comparison of mitochondrial genes could indicate that the nocturnal geckos formed a distinct genetic group (lines 9–10).

In your answer, explain how new techniques enable the comparison of genes to be completed rapidly.

[3 marks]

Question 10 continues on the next page

1 0 . 5

Describe and explain **two** precautions required to ensure that the estimate of the size of the population of geckos was valid (lines 12–13).

Do **not** include sample size as one of the required precautions.

In your answer, include the formula to estimate the size of the population using the mark-release-recapture method.

[3 marks]

Precaution 1 _____

Precaution 2 _____

Formula _____

1

END OF QUESTIONS

[illegible]

Question number	Additional page, if required. Write the question numbers in the left-hand margin.

